Denial Appeals AI

Sprint Timeline – April 6, 2026 *(stories in execution order — companion to the Workstream Alignment doc)*

Project Objective

Building an AI-powered denial appeals management system that integrates with the customer's existing Azure AI Foundry instance to ingest denial documents, gather supporting evidence, generate appeal letters via the existing letter writer agent, route them through human-in-the-loop review, and deliver approved letters — accelerating appeal turnaround time and increasing overturn rates while maintaining full audit and compliance traceability.

Sprint Timeline Overview

This document sequences the 39 stories from the backlog across 6 two-week sprints (April 6 – June 26, 2026). Each sprint has a clear theme, goal, and a balanced mix of stories drawn from multiple workstreams. The same backlog organized by capability is in **WORKSTREAMS.pdf**; this document is the time-ordered companion view used for sprint planning and stakeholder updates.

Sprint 1	Sprint 2	Sprint 3	Sprint 4	Sprint 5	Sprint 6
Apr 6 – Apr 17	Apr 20 – May 1	May 4 – May 15	May 18 – May 29	Jun 1 – Jun 12	Jun 15 – Jun 26

32	173	7	~29	12
STORIES COMMITTED	POINTS COMMITTED	BACKLOG STORIES	AVG PTS / SPRINT	CALENDAR WEEKS

WS1 Infrastructure · WS2 Security · WS3 Backend · WS4 Frontend · WS5 Workflow / AI · WS6 Quality

Sprint 1 — Foundation

April 6 – April 17 · 5 stories · 26 pts

Goal: Stand up Azure infrastructure in the customer tenant, lock down credentials, and establish CI/CD so the team is unblocked. Begin the Entra ID app registration in parallel since customer IT approval can take time. By the end of this sprint, the empty React shell is deployed and accessible via the Azure Static Web App URL.

ID	WS	Story	Pts	Priority
FV-001	WS1	**Provision Azure Resources (RG, SWA, Key Vault, App Insights)**	5	Highest
FV-005	WS1	**Terraform / Bicep Modules for All Resources**	8	High
FV-002	WS1	**Azure DevOps CI/CD Pipeline**	5	Highest
FV-010	WS2	**Azure AD App Registration & OAuth 2.0 Setup**	5	Highest
FV-013	WS2	**Security Audit — Remove Hardcoded Credentials**	3	Highest

Sprint 2 — Security & API Proxy

April 20 – May 1 · 5 stories · 24 pts

Goal: Get authentication flowing end-to-end through MSAL.js, deploy the Azure Functions API proxy that bridges the frontend to Foundry, and stand up separate staging and production environments. By the end of this sprint, a developer can sign in via Entra ID, the proxy is live, and a request from the SPA reaches Foundry through the full pipeline.

ID	WS	Story	Pts	Priority
FV-011	WS2	**Integrate MSAL.js in React Frontend**	5	**Highest**
FV-020	WS3	**Create Azure Functions API Proxy**	8	**Highest**
FV-021	WS3	**Proxy — Conversations API Support**	5	**High**
FV-022	WS3	**Proxy — Rate Limiting & Request Validation**	3	**High**
FV-003	WS1	**Provision Staging and Production Environments**	3	**High**

Sprint 3 — Core Features

May 4 – May 15 · 5 stories · 32 pts

Goal: Replace localStorage with server-side persistence, implement real-time workflow execution tracking against Foundry, expand the YAML parser to cover all action types, and stand up the server-backed HITL queue with WebSocket notifications. This is the heaviest sprint by points because it lays the foundation for the appeals workflow in Sprint 4.

ID	WS	Story	Pts	Priority
FV-030	WS3	**Replace localStorage with Server-Side Persistence (Cosmos DB)**	8	**High**
FV-031	WS4	**Error Boundary & Global Error Handling**	3	**High**
FV-040	WS5	**Real-Time Workflow Execution Tracking**	8	**High**
FV-041	WS5	**Workflow YAML Parser — Full Action Support**	5	**High**
FV-050	WS5	**HITL Queue — Server-Side with WebSocket Notifications**	8	**High**

Sprint 4 — Denial Appeals Workflow

May 18 – May 29 · 5 stories · 29 pts

Goal: Wire up the denial appeals workflow end-to-end. Build the appeals backend API, integrate with the customer's existing letter writer agent via the Foundry workflow, implement the multi-step HITL conversation view, and add letter export to PDF/DOCX. By the end of this sprint, a reviewer can process a real denial appeal from intake through approved letter export.

ID	WS	Story	Pts	Priority
FV-060	WS3	**Denial Appeals Backend API and Data Model**	8	**High**
FV-061	WS5	**Denial Appeals — Foundry Workflow Integration**	8	**High**
FV-062	WS5	**Appeal Letter Generation and Export (PDF/DOCX)**	5	**High**
FV-051	WS5	**HITL Queue — Multi-Step Conversation View**	5	**High**
FV-012	WS2	**Role-Based Access Control (Admin / Reviewer / Viewer)**	3	**High**

Sprint 5 — Polish & Monitoring

June 1 – June 12 · 6 stories · 26 pts

Goal: Add the production-grade hardening that turns the working app into a polished product. Front Door + custom domain for the public face, Application Insights telemetry, operations dashboard, audit logging for compliance, and frontend performance optimizations (code splitting, skeleton loading). The team begins to slow new feature work and shift focus toward stability and operational readiness.

ID	WS	Story	Pts	Priority
FV-004	WS1	**Configure Azure Front Door with Custom Domain**	5	**High**
FV-070	WS6	**Application Insights — Frontend Telemetry**	3	**High**
FV-071	WS6	**Operations Dashboard in Azure Monitor**	5	**High**
FV-072	WS2	**Audit Logging (7-year retention)**	5	**High**
FV-034	WS4	**Code Splitting and Performance Optimization**	5	Medium
FV-032	WS4	**Loading States and Skeleton Screens**	3	Medium

Sprint 6 — Testing & Launch

June 15 – June 26 · 6 stories · 36 pts

Goal: Comprehensive test coverage, accessibility audit, responsive design for tablets, in-app notifications, and load testing. By the end of this sprint, the application is test-covered, accessible, and validated under load — ready for production launch and the start of the pilot rollout.

ID	WS	Story	Pts	Priority
FV-080	WS6	**Unit Test Suite (Vitest + RTL)**	8	**High**
FV-081	WS6	**End-to-End Test Suite (Playwright)**	8	**High**
FV-082	WS6	**Load Testing (50 / 100 concurrent users)**	5	Medium
FV-090	WS4	**Keyboard Navigation and Screen Reader Support**	5	Medium
FV-033	WS4	**Responsive Design — Tablet and Mobile Support**	5	Medium
FV-091	WS4	**Notification System (in-app + browser push)**	5	Medium

Stretch / Post-Engagement Backlog

After June 26 · 7 stories · 31 pts

Goal: These stories are intentionally deferred to fit the 6-sprint, 12-week timeline within team capacity. They are high-value but not critical for the production launch. They should be pulled into Sprint 7+ after the engagement based on customer feedback and adoption signals from the pilot. Several are good candidates to bundle into a "Phase 2 expansion" follow-on engagement.

ID	WS	Story	Pts	Priority
FV-042	WS5	**Agent Detail Panel — Full Configuration View**	3	Medium
FV-052	WS5	**HITL Queue — Assignment and Routing**	5	Medium
FV-064	WS5	**Denial Appeals — Dashboard and Analytics**	5	Medium
FV-065	WS5	**Denial Appeals — Evidence Upload and Management**	5	Medium
FV-063	WS5	**Denial Appeals — Bulk Processing**	5	Medium
FV-043	WS5	**Workflow Comparison View**	5	Low
FV-092	WS4	**Onboarding and Help System**	3	Low

Capacity, Velocity & Notes

- **Sprint sizing assumes a 4-engineer team with ~32 pts/sprint sustainable velocity.** The committed scope (173 pts across 6 sprints) leaves a small buffer for refinement, regression fixes, and unplanned work. Sprints 3 and 6 are intentionally heavier (32 and 36 pts) because they contain the most critical features for the production pilot.
- **Sprint 1 is 100% Infrastructure & Security workstreams** by design — nothing else can begin until the foundation is in place. Customer Azure access provisioning is the biggest external risk and must be initiated before Sprint 1 day 1.
- **Sprint 4 is the production pilot milestone.** By end of Sprint 4 the customer has a working denial appeals workflow they can run with real data. Sprints 5 and 6 are about hardening, not new features. This staging makes a customer demo at the end of Sprint 4 the natural "show and tell" point.
- **Workstream balance per sprint** is intentional. No single workstream dominates a sprint; each sprint draws from multiple workstreams so different team members are productive in parallel. Sprint 3 has 3 stories from WS5 (Workflow / AI) only because they are the foundational pieces other features depend on.
- **The 7-story backlog (31 pts) represents ~1 additional sprint of work.** If team velocity exceeds the 32 pts/sprint assumption, these can be pulled forward; if it falls short, they stay deferred without putting the production launch at risk. They are explicitly NOT on the critical path.
- **Bi-weekly sprint demos to the customer** are scheduled for Friday of each sprint's second week (Apr 17, May 1, May 15, May 29, Jun 12, Jun 26). Customer attendance is the single best mechanism for catching scope drift early.

** Sprint dates assume Monday start, Friday end of week 2, and exclude US holidays. Memorial Day (May 25) falls in Sprint 4 — capacity is reduced by 1 day for that sprint.*

*** Story points are initial estimates and should be refined by the team during refinement sessions before sprint commitment. Adjustments will be reflected in the next status update.*

**** Sprint assignments may shift based on dependency discoveries during sprint 1 and customer feedback after each sprint demo. Backlog stories are candidates for promotion if velocity allows.*